EFY TESTED
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Part 2

ANTI-COLLISION SYSTEM For Vehicles

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Part 1 of this article published in June issue (which is also available at *https://www.electronicsforu.com/electronics-projects/lidar-based-anti-collision-system*) covered the making of a lidar based anti-collison alert system for vehicles with Raspberry Pi. In this Part 2 we will carry forward the design for the detection of cars and road lanes for an advanced driver assistance system (ADAS) using an automotive-grade system basis chip (SBC), or NVIDIA Jetson.

If you are making this project just for learning, you can even use a Raspberry Pi with 2GB or 4GB RAM. So, in addition to the components used in the previous project, you will need the components listed in the bill of material table here. Fig. 1 shows the author's prototype with its camera attached to the SBC. Fig. 2 shows the testing of the lane detecting project's prototype.

The project requires 5V, 4A power supply, which can be obtained from the vehicle itself. But for testing purpose, you can power the NVIDIA Jetson development kit with 5V, 2A USB supply. You may

BILL OF MATERIAL

Components	Quantity	Description
NVIDIA Jetson/ Raspberry Pi	1	4GB RAM
Webcam/Raspberry Pi cam	1	For night vision (5 to 10MP)
SD card	1	32/64GB
HDMI 17.8cm (7-inch) display	1	HDMI display

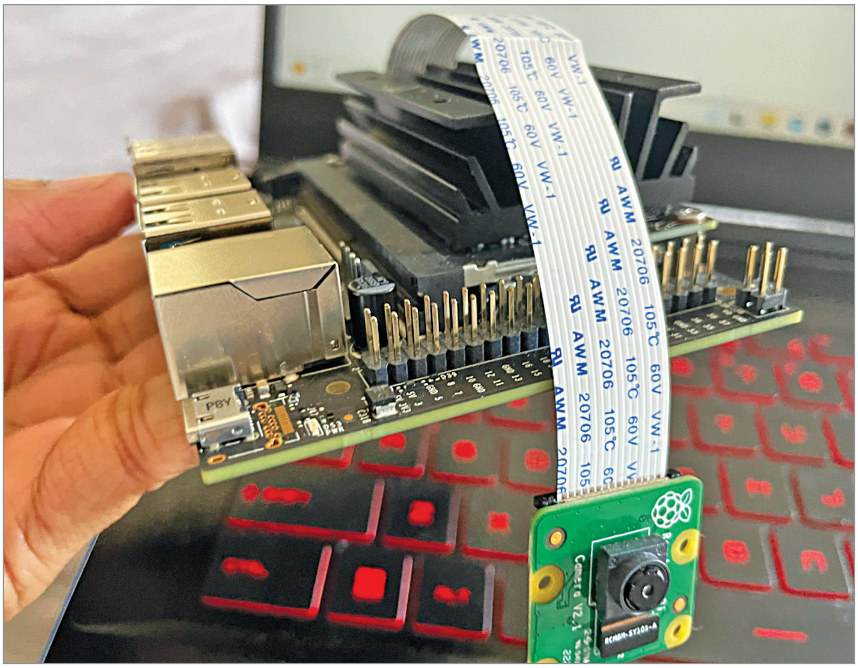

Fig. 1: Camera attached to SBC

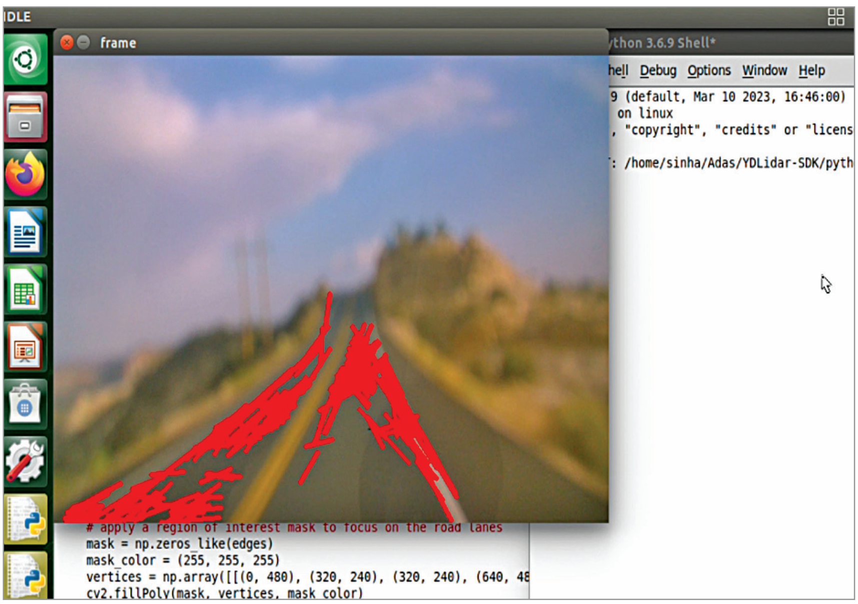

Fig. 2: Testing of the lane detecting prototype